

3rd

```
GNU nano 8.7.1 fork_demo.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child Process\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());

        sleep(10);
    }
    else
    {
        printf("Parent Process\n");
        printf("PID : %d\n", getpid());

        wait(NULL);
    }

    return 0;
}
```

OUTPUT :

```
abhis@MIKHAIL-37:~$ nano fork_demo.c
abhis@MIKHAIL-37:~$ gcc fork_demo.c -o fork_demo
abhis@MIKHAIL-37:~$ ./fork_demo
Parent Process
PID : 676
Child Process
PID : 677
PPID : 676
```

4th

```
GNU nano 8.7.1 multi_fork.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int i;

    for(i=0; i<3; i++)
    {
        if(fork() == 0)
        {
            printf("Child %d PID=%d\n", i+1, getpid());
            return 0;
        }
    }

    for(i=0; i<3; i++)
    {
        wait(NULL);
    }

    printf("All children completed.\n");

    return 0;
}
```

OUTPUT :

```
abhis@MIKHAIL-37:~$ nano multi_fork.c
abhis@MIKHAIL-37:~$ gcc multi_fork.c -o multi_fork
abhis@MIKHAIL-37:~$ ./multi_fork
Child 1 PID=734
Child 2 PID=735
Child 3 PID=736
All children completed.
```